

Reasoning Tokens Buy Competence, Not Compliance: What Test-Time Compute Does and Does Not Purchase, Measured with a Sign-Exact Oracle

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Abstract

Test-time reasoning is usually evaluated on one axis: does more thinking buy more accuracy per token? We add a second axis that efficiency work has ignored, *compliance*: when an explicit instruction opposes a convention the model absorbed in training, do reasoning tokens help the instruction win? Using CircuChain v2, a circuit-analysis probe whose answers are verifiable to the sign by four independent solvers, we measure both axes across 18 model configurations from 7 families. Three findings. (1) **Reasoning buys competence only**: toggling the same weights between reasoning and non-reasoning modes at five scales (Qwen3 1.7B–32B) raises magnitude accuracy at every scale (from $1.7\times$ at 32B to $12\times$ at 1.7B) while reversion to the trained convention stays within a narrow band: 63–81% (configuration mean 74.2%; pooled over variables 75.9%), a band that also contains GPT-5, Gemma-4, GPT-OSS-120B, and DeepSeek-V3 across a $792\times$ competence range. (2) **The GPT-5 reasoning-effort ladder dissociates the axes**: at minimal effort it answers from the prior, cheaply and wrongly; at low effort it solves sign-trap circuits to nine decimals yet still reports the trained frame $\sim 71\%$ of the time; at default effort reasoning consumes the entire token budget and *zero* of 121 requests return an answer, at the highest cost per instance. (3) **Unbounded rumination is a real cost**: at 4B–32B, reasoning modes burn the full 16k budget on 73–77% of problems without converging (the 1.7B model instead fails fast, truncating on only 16%). Across all models, flipping one convention instruction breaks 401 previously-solved problems and fixes 6. Efficiency evaluations that report accuracy-per-token implicitly assume compute also purchases controllability; on this benchmark, at every budget we tested, it does not. The generator, oracle, and harness will be released; total study cost was under \$100.

1 Introduction

The efficient-reasoning agenda optimizes a ratio: capability delivered per unit of test-time compute (Snell et al., 2024; Muennighoff et al., 2025). Its numerator is almost always *accuracy*. This paper measures a second numerator that practitioners implicitly assume comes bundled with the first: *compliance*, whether the model does the task *the way it was told to* when the instruction opposes a strong prior from training. Instruction-following suites score constraints that do not oppose such priors (Zhou et al., 2023); recent work names the opposed case (Zhang et al., 2025; Wu et al., 2024) but scores it with rubrics or judges rather than an exact oracle, and prior-following failures surface more broadly as sycophancy (Sharma et al., 2023).

The distinction matters exactly when models are deployed against specifications: house styles, legacy code conventions, discipline-specific sign disciplines, safety policies that differ from common practice. If reasoning tokens buy accuracy but not obedience, then (i) scaling test-time compute does not close the steerability gap, (ii) efficiency benchmarks that score only correctness will not see the gap at any budget, and (iii) budget tuning (effort levels, early stopping, distillation) can be evaluated on a false frontier.

Measuring this cleanly requires a task where compliance is *objectively decidable at the precision the instruction affects*. We use linear DC circuit analysis: the answer is exact; a sign convention (e.g. the direction mesh currents are defined) changes only the *reporting frame*, never the physics; and the textbook prior is overwhelmingly one-sided (clockwise). CircuChain v2 poses 1000 instances (125 four-way-verified physics configurations $\times$ {default contract, three single-factor convention flips} $\times$ {mesh, nodal}) and grades deterministically against an oracle validated independently of the transform code (125/125 configurations). Within-physics matched pairs give exact McNemar tests (McNemar, 1947): the causal effect of one changed sentence.

Contributions for this workshop. (1) A compute-vs-compliance measurement: the reasoning toggle held at fixed weights across five scales, plus the GPT-5 effort ladder, with per-instance dollar costs. (2) The negative result: no reasoning budget we tested, at any scale or effort, measurably reduces reversion to the trained convention (63–81% across all configurations). (3) An efficiency pathology inventory measured with an exact oracle: prior-driven fast-wrong answers at low effort, budget-consuming rumination at high effort, and 100%-truncation regimes where the most expensive configuration returns nothing. (4) A seeded, LLM-free instrument (to be released) cheap enough for budget-sweep studies (full panel $< \$100$).

2 Instrument in brief

CircuChain v2 extends the published CircuChain probe (Ravishankara, 2026a) and shares its deterministic, generator-anchored design philosophy with checkpointed engineering-plot evaluation (Ravishankara, 2026b); Ravishankara and Persad Maharaj (2025) surveys this broader move toward reasoning-centric evaluation. 125 physics configurations across seven linear-DC topologies are accepted only when two independent hand-derived solvers, an exact rational-arithmetic MNA solver, and NGSPICE agree on every quantity; sign traps and controls are balanced $\sim 1:1$ by code predicates. Each physics is posed under the textbook-default contract and three single-factor flips (mesh direction $\text{cw} \rightarrow \text{ccw}$, passive $\rightarrow$ active sign, reference node bottom $\rightarrow$ top), in both mesh (KVL) and nodal (KCL) methods. Expected answers in every frame are validated independently of the transform code, including a *physically re-grounded* SPICE realization of the reference-node flip. Grading is deterministic (structured answer-line parser; magnitude at 2%; sign compliance in the *instructed* frame; truncation recorded as an outcome). Decoding: temperature 0.6, top- p 0.95, 16,384 completion budget for reasoning modes, 4,096 otherwise. Variables whose sign differs between instructed and default frames are *diagnostic*; reversion is measured on those, given correct magnitude. Whole-problem correctness and variable-level frame choice are distinct estimands: the former measures end-to-end controllability but is informative only where a model can solve the circuit; the latter conditions on local magnitude success and is measurable across the entire panel. We keep the two separate throughout.

3 Reasoning raises competence, never compliance

Qwen3’s native /think toggle switches reasoning on/off with weights, prompt, serving stack, and decoding policy fixed: the cleanest available compute contrast. Table 1 shows the five-scale result: magnitude competence rises with reasoning at every scale, from $1.7\times$ at 32B to $12\times$ at 1.7B (14B: 2.8% $\rightarrow$ 9.3%, a $3.3\times$ gain), while reversion to the clockwise frame among magnitude-correct diagnostic variables stays within 68–81%, with overlapping Wilson intervals (Wilson, 1927) at every scale. Reasoning tokens purchase physics, not obedience.

The cost side. Table 2 shows what the reasoning budget buys at the small end: below the competence threshold, think modes at 4B–32B ruminate to the full 16k budget on most problems (truncations: 727–769 of 1000); these are genuine self-correction spirals that never converge, the overthinking phenomenology of R1-style reasoning models (DeepSeek-AI, 2025; Chen et al., 2024) measured here with an exact oracle. The 1.7B model is the exception:

Scale	No-think		Think (16k budget)	
	Mag. correct	Reversion (n)	Mag. correct	Reversion (n)
1.7b	0.2%	71% (24)	2.4%	72% (69)
4b	0.7%	68% (37)	5.0%	70% (47)
8b	1.2%	70% (61)	2.1%	81% (31)
14b	2.8%	75% (91)	9.3%	78% (91)
32b	5.1%	77% (102)	8.6%	76% (62)

Table 1: The reasoning toggle at fixed weights (Qwen3 ladder). Reversion = share of magnitude-correct sign-diagnostic variables reported in the trained clockwise frame despite the counter-clockwise instruction.

Scale	No-think		Think	
	Mag. correct	Truncated	Mag. correct	Truncated
1.7b	0.2%	30	2.4%	164
4b	0.7%	1	5.0%	742
8b	1.2%	1	2.1%	732
14b	2.8%	0	9.3%	727
32b	5.1%	4	8.6%	769

Table 2: Rumination economics on the Qwen3 ladder (of 1000 instances). At 4B and above, reasoning modes burn the full completion budget on most instances while magnitude competence stays under 10%; the 1.7B model instead fails fast.

it truncates on only 164 of 1000, failing quickly rather than ruminating. This is paid-for compute delivering neither competence nor compliance; larger budgets shift the wall rather than remove it.

4 The GPT-5 effort ladder: three budgets, three failure modes

GPT-5’s reasoning-effort control exposes the compute–behavior trade directly on one frontier model (Figure 1):

minimal ($\approx 1.6\epsilon$ /instance): reasoning tokens = 0; the model answers instantly, *from the prior*, and gets the physics wrong on our probe (archived with the artifacts). **low** ($\approx 7.4\epsilon$): on a 248-instance run it solves sign-trap circuits exactly when it completes (magnitude-correct 32.7% overall), and still reports the trained frame on 70.6% of magnitude-correct diagnostic variables; the paired ccw flip breaks 10 problems and fixes 0. **default** ($\approx 8.3\epsilon$, the most expensive): hidden reasoning consumed the *entire* 8,192-token budget on 121 of 121 requests; zero answers returned, at the highest per-instance cost (the raw run is archived with the artifacts).

The ladder dissociates the axes cleanly: compliance is not purchasable at any of the three budgets, competence is purchasable at exactly one of them, and the default configuration purchases nothing. For efficiency work, the default row is the cautionary tale: without budget-aware evaluation, a deployment can pay maximum cost for a 0% answer rate.

5 A stable rate that compute does not move

Figure 2 is the study’s central object: reversion to the trained clockwise frame among magnitude-correct diagnostic variables, for every panel configuration with $n \geq 20$, against task competence on a log axis. The band is 63–81% (configuration mean 74.2%; pooled over variables 75.9%) across 18 configurations spanning 7 families, $\sim 792 \times$ in competence, two serving stacks, and both reasoning modes. A weighted logistic regression of reversion on log competence finds no significant trend ($z = 1.40$, $p = 0.16$): the data are consistent with a stable rate across the measured range, though several configurations contribute small diagnostic denominators, so we describe the band as broadly stable rather than proven

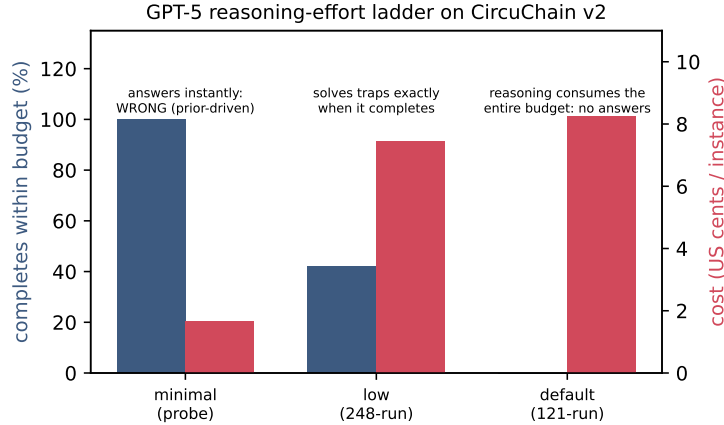


Figure 1: GPT-5 reasoning-effort ladder on CircuChain v2: completion-within- budget rate (left axis) and measured-effort cost per instance (right axis). minimal completes always but answers from the prior (wrong); low is the only configuration that buys competence; default spends the most and returns nothing. At the low setting, the only one that both reasons and answers, compliance with the counter-normative convention is $\sim 29\%$.

invariant. Aggregated over all models, the within-physics paired flip breaks 401 solved problems and fixes 6 (67:1). This asymmetry alone establishes that the flip causally breaks otherwise-solved problems; the *reversion-to-prior* reading follows from its conjunction with the controls below.

Three controls establish that this is convention-specific reversion rather than generic instruction decay: the same models follow matched *non-convention* unusual instructions (reversed answer order, scientific notation) at 85–97%; GPT-5 and Gemma obey the *other two* convention factors while disobeying mesh direction; and the collapse is similarly large on trap and control physics (Gemma: $0.77 \rightarrow 0.10$ vs. $0.87 \rightarrow 0.13$). The reversion also persists under nodal solving, where loop orientation never enters the derivation; it is a property of the *reporting frame*, not the procedure. Resampling ($k=5$) of the most competent model shows a stable rate (SD 4.7pp) over stochastic items: frame choice behaves like a biased coin, roughly 3:1 toward the trained frame study-wide (6:1 in the resampled Gemma subset), rather than a fixed property of particular items.

6 Implications for efficient reasoning

1. Put compliance in the efficiency numerator. Accuracy-per-token frontiers implicitly assume behavior is purchasable with compute. Measured with an exact oracle, we find no such purchase: the compliance curve shows no measurable improvement with compute in our panel. Efficiency evaluations (especially of effort controls, early-stopping, and distillation) should report *constraint adherence per token* alongside accuracy per token, using probes whose compliance is mechanically decidable (ours will be released; it costs cents per thousand instances).

2. Budget-aware evaluation is not optional. The most expensive GPT-5 configuration returned zero answers; small reasoning models at 4B–32B burn full budgets on 73–77% of instances without converging. Truncation must be a first-class reported outcome, or efficiency comparisons silently reward configurations that never finish.

3. Effort controls are behavioral, not just economic. Moving GPT-5 from minimal to low switches it from prior-recitation to genuine derivation, a qualitative regime change purchased with tokens; past that point, additional tokens purchase nothing we can measure. Efficiency work that tunes effort levels is implicitly selecting *which* failure mode ships.

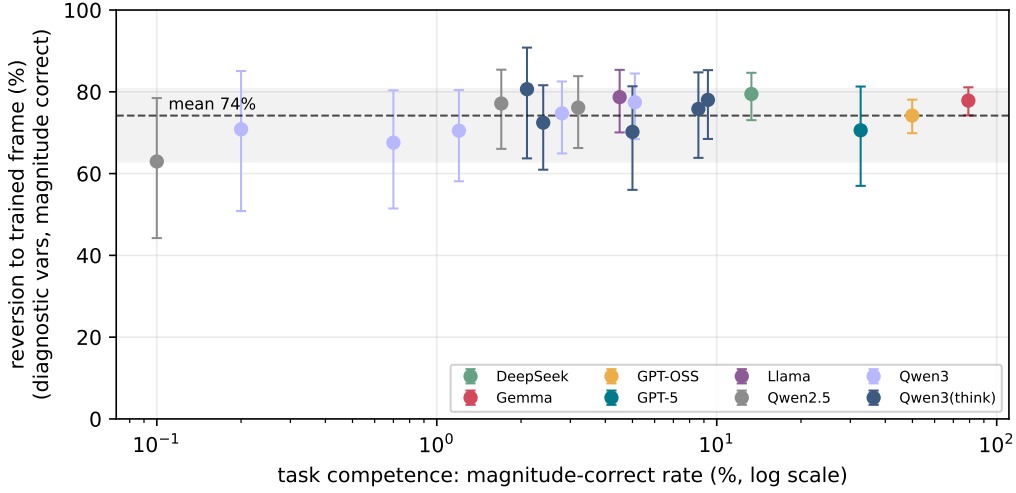


Figure 2: Reversion to the clockwise prior is broadly stable across competence, scale, family, serving stack, and reasoning mode. Each point is one panel configuration (Wilson 95% CIs; $n \geq 20$ diagnostic variables); shaded band = observed range 63–81%.

4. Distillation and steering research have a concrete target. The frame choice is made outside the reasoning trace (it survives nodal solving and every budget we ran), and it is sampled (roughly 3:1) rather than committed. A cheap deterministic frame-checking/repair layer recovers more reliability per dollar than any reasoning budget we measured.

7 Limitations

One domain (linear DC circuits); a second convention domain (handedness / cross products) is in progress. Whole-problem effects rest on the few models competent at multi-loop circuits (itself an efficiency datum: 70B-class instruct models sit under 14% magnitude-correct); variable-level results cover the full panel. GPT-5 was run at 248 instances under a bounded budget; its numbers are conservative. Single-sample decoding for most columns is mitigated by the resampling study and the cross-column invariance.

Reproducibility

Seeded byte-reproducible generator (125 physics), four-way oracle, independent transform validation (125/125), deterministic grading, full serving provenance per response, canary-stamped evaluation subset; code and harness to be released upon publication. Total API cost < \$100; local runs on one Apple M5 Max.

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